

*Teaching Notes 8A**

Beyond MM: Agency Conflicts

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Overview

- Focus on fourth MM assumption
 - Capital Structure does not affect investment policy and cash-flows
- Relaxing this assumption implies that there are “agency” conflicts between debt and equity
- Leads to a more complete view on how firms choose capital structure

Objectives

- Understand the “qualitative” aspects that are introduced when we relax MM assumptions
- Understand “quantitatively” what relaxing a particular assumption implies for capital structure

Assumption #4

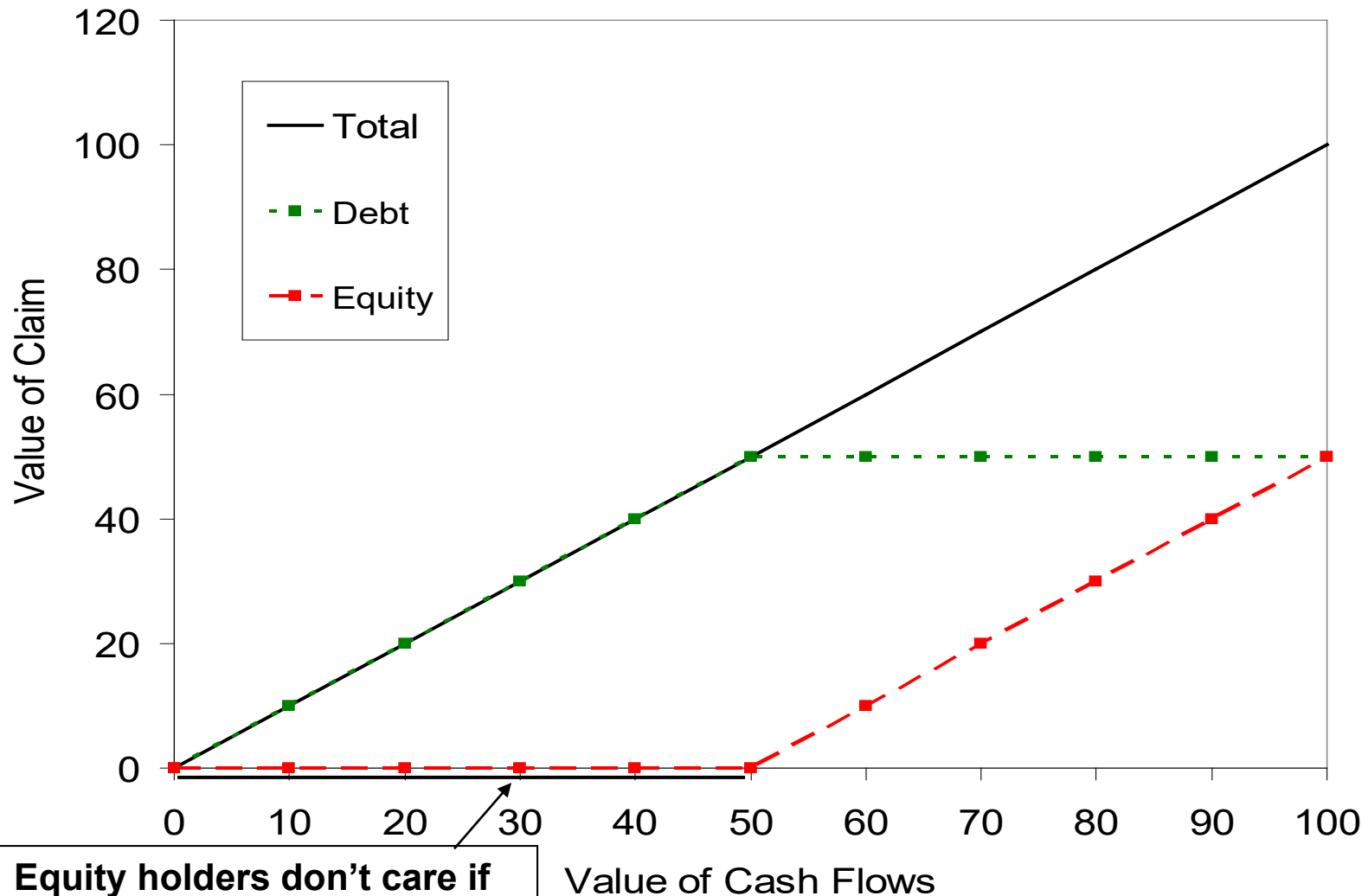
Capital Structure does not Impact
Investment Policy and Cash Flows

(aka “No Agency Conflicts”)

Root Cause of Assumption Failing: Conflicts between Debt and Equity Holders

- The root cause for why capital structure affects investment policy/ CFs
 - Conflicts between debt and equity holders
 - To see this, let us look at their payoffs
- **Equity Payoff:**
 - Holding a **call option** on the assets of the firm
- **Debt Payoff:**
 - Owning firm's assets, but being **short a call option**
 - Equity holders can “call” the firm by making interest payments
- Different payoff structures mean equity and debt holders may have conflicting interests

Root Cause of Assumption Failing: Conflicts between Debt and Equity Holders



**Equity holders don't care if
bigger losses occur once
firm is in this range**

Three Types of Agency Conflicts

Three Main Conflicts

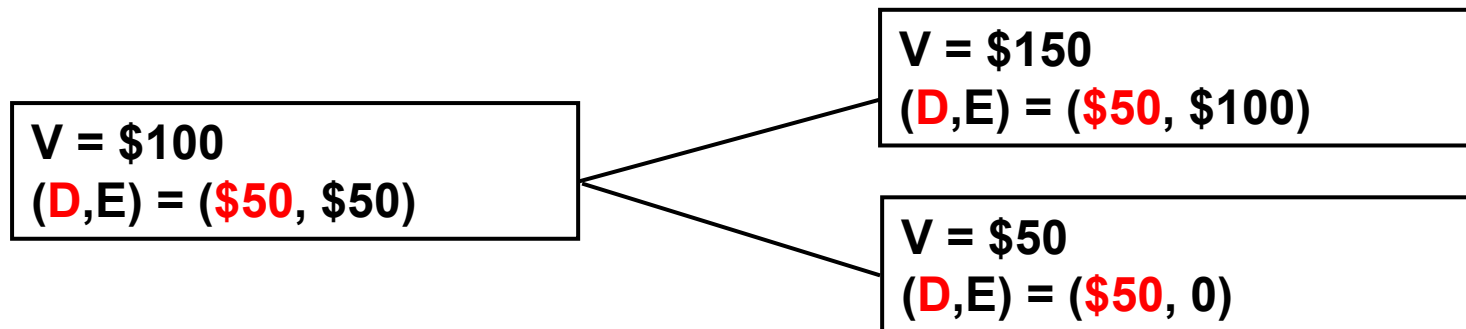
1. Wealth transfers from debt-holders to shareholders (“cashing out”)
2. Risk shifting (“asset substitution”)
 - Equity holders’ option is worth more when cash flows are more volatile
3. Debt overhang (“underinvestment problem”)
 - Can’t raise new equity if current debt likely to be underwater
 - All problems emerge if debt is “risky”

Wealth Transfer

Example 1: Wealth Transfer

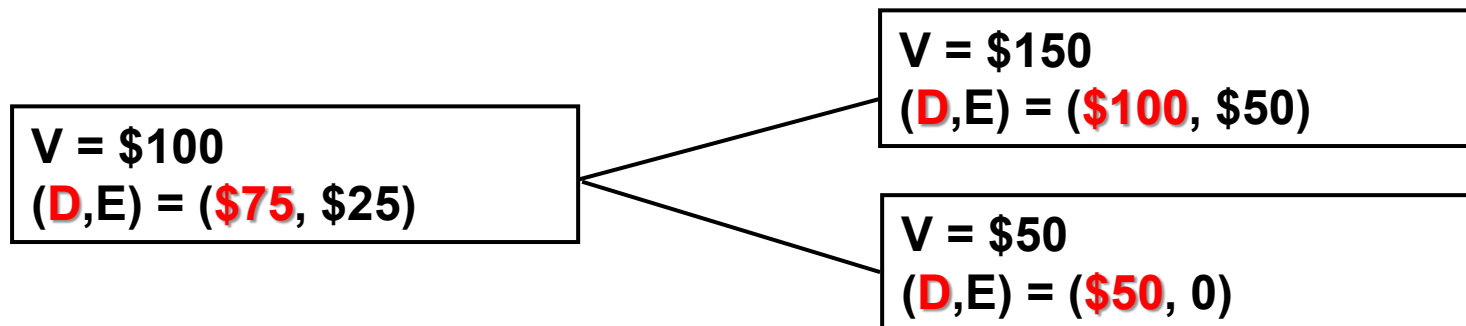
- Basic Idea – **“Stealing”**
 - Illustrate with an example
- Two periods: $t = 0$ and $t = 1$
 - Assume $r_f = 0$ and $\beta = 0$ to keep it simple
 - Logic the same with positive interest rates and risk
- At $t = 1$, the state is either boom or bust, with equal probability
 - Value of firm goes to 150 in boom, down to 50 in bust
- Firm has debt with face value of 50 due at $t = 1$
 - Use D and E to represent market values
- Firm has value of 100 at $t = 0$.

Example 1 (contd.): Wealth Transfer



Note: Debt is not risky so far. Why? Payments equal to face value of debt made in all states

→ → Now issue more debt with face value of 50 and equal seniority to existing debt (also called “pari passu” debt), repurchase shares:



Example 1 (contd.): Wealth Transfer

- Value of old debt post-issue?
- Value of new debt?
- Value of equity post-issue?
- Value of equity post-issue plus payment to shareholders?
- Who won and who lost?
- Was value of total claims maintained?

Example 1 (contd.): Wealth Transfer

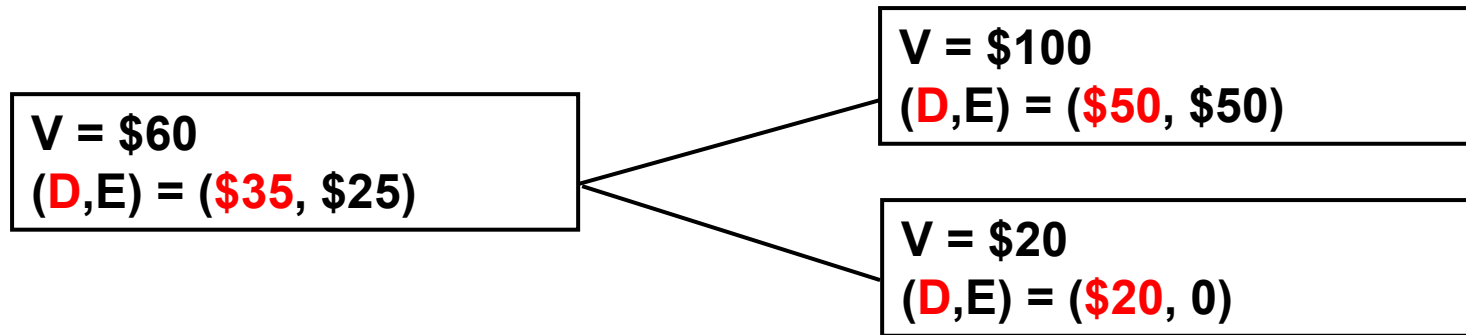
- Lesson: Additional debt issue dilutes existing debt holders to the benefit of shareholders
 - Total Debt became “risky”
 - Debt payments in each state not equal to total face value of debt
- More blatant wealth transfers are possible
 - Issue new debt or sell assets to pay a dividend to shareholders
- Wealth Transfer in Real Life:
 - Bank bailout (see Scharfstein & Stein article, and Veronesi and Zingales (2010) paper)

Risk Shifting

Example 2: Risk Shifting

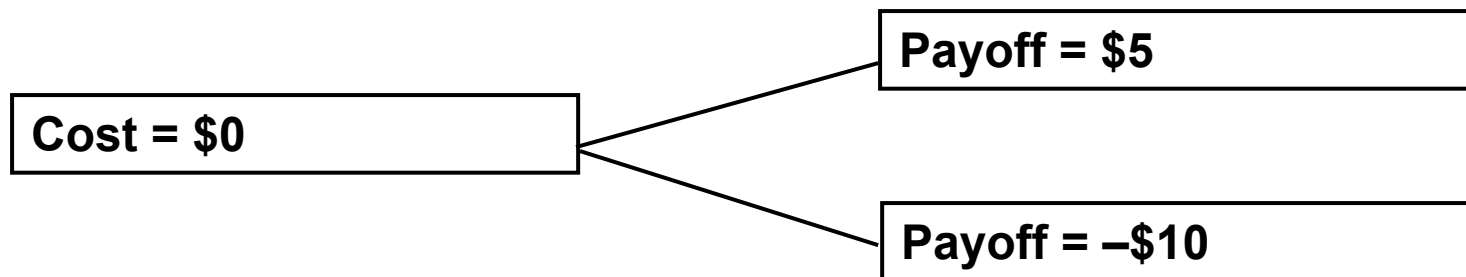
- Basic idea: equity holders' option on the firm is worth more when cash flows are more volatile
 - Management willing to take negative NPV projects that “introduce” more volatility
 - Shareholder receive all the upside and none of the downside: shift the downside risk to “someone else” while keeping the upside
- If management is close to bankruptcy, sell all of the assets and go to Las Vegas
 - Blatant transfers and theft may be easy to detect, but “risk shifting” is less obvious

Example 2 (contd.): Risk Shifting



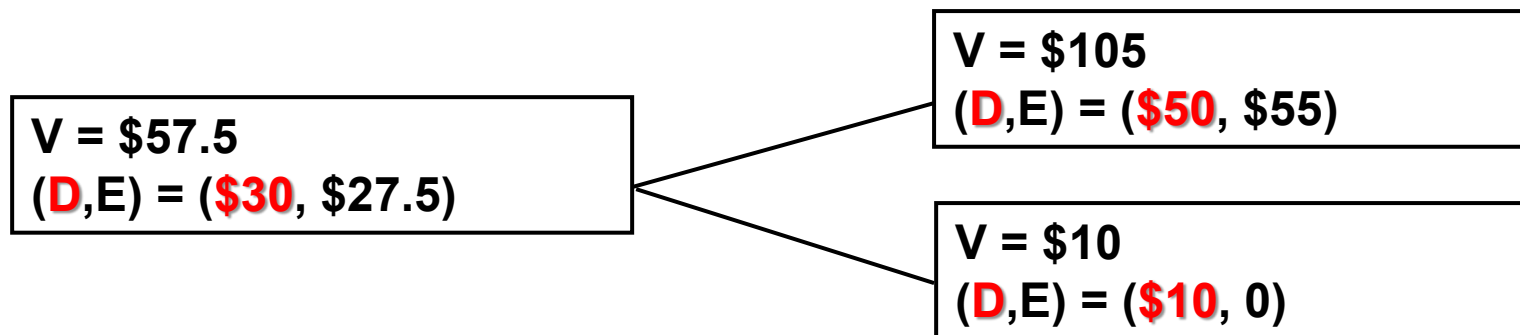
Note: Debt is already “risky”. Why? Payments in all states of the world do not equal face value of debt

Suppose a new investment opportunity arrives...



- $\text{NPV} = -\$2.5$

Example 2 (contd.): Risk Shifting



- Investment transfers \$5 from D and increases E by \$2.5!
 - So do shareholders want to undertake the project?
- Feelings of initial creditors?
 - Does MM hold?

Example 2 (contd.): Risk Shifting

- Risk shifting arises when equity holders benefit from increasing the risk of the project
 - Option analogy: Equity benefits from volatility
- Involves transfers from “bad” state (where D is marginal claimant) to “good” state (where E is marginal claimant)
- Only arises when D is not always fully paid
 - Requires risky debt

Example 2 (contd.): Risk Shifting

- Example: Savings and Loan (S&L) crisis in the 1980s
 - S&Ls are banks with primarily mortgage loans as assets and primarily deposits as liabilities
 - Early 1980s saw a combination of bad conditions for these banks
 - Many S&Ls took on risky loans that would destroy capital if they went badly, but would save the firm if they paid off
 - See article on Canvas
- Example 2: Literal “Gambling for Resurrection” (Fedex 1970s)

BUSINESS 15/10/2012 16:19 BST | Updated 15/10/2012 16:21 BST

Fred Smith, FedEx Founder And CEO, Once Gambled \$5,000 On Blackjack To Keep Company Alive



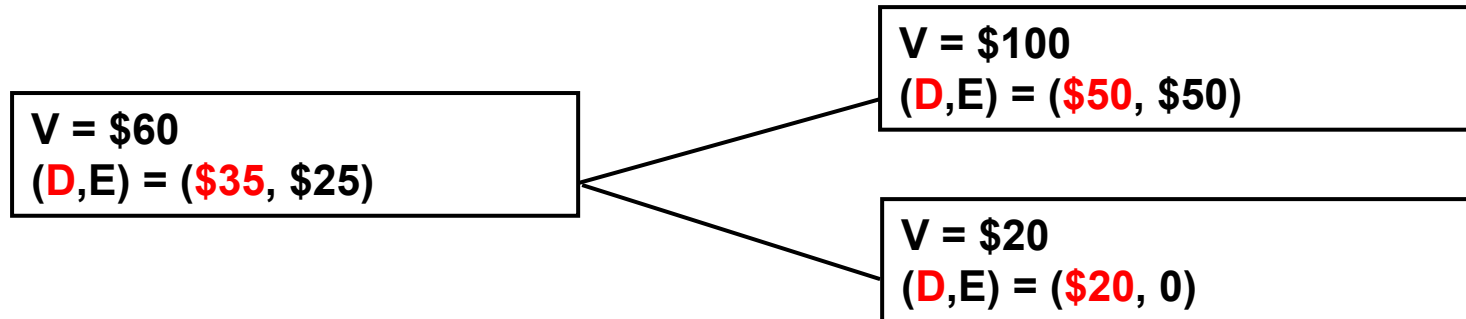
By Harry Bradford, HuffPost US

Debt Overhang

Example 3: Debt Overhang

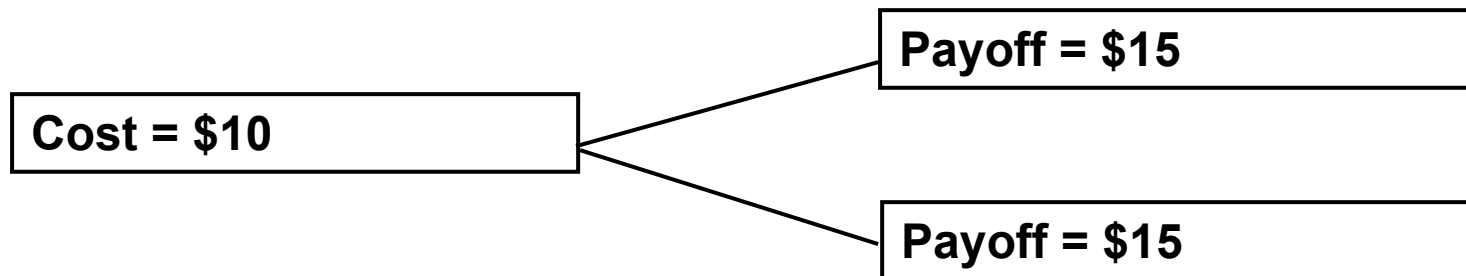
- Basic Idea: if a firm's existing debt is underwater in some states, then it may not be able to finance +NPV projects with equity
 - New investors reluctant to invest because some of the returns will go to pay existing creditors
- Debt overhang leads to underinvestment

Example 3 (contd.): Debt Overhang



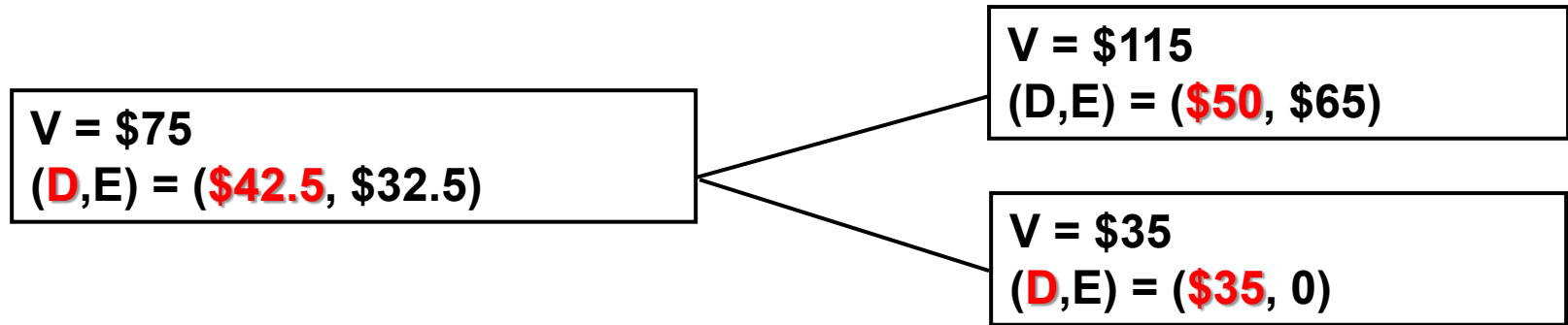
Note: Debt is already “risky”. Why? Payments in all states of the world do not equal face value of debt

Suppose a new investment opportunity arrives...



- NPV = \$5

Example 3 (contd.): Debt Overhang



- This investment raises the value of D and E at $t=0$ by \$7.5 each
 - So can we raise \$10 of new equity (or debt)?
- Solutions?

Example 3 (contd.): Debt Overhang

- Lesson: debt introduces distortions that might make it hard to raise additional equity financing in the future
 - Even when projects are profitable!
- This partly explains why growth companies don't use debt
 - Provides rationale for why Chapter 11 allows DIP (Debtor In Possession) financing to be senior to existing creditors
- Debt Overhang in Real Life
 - See articles on Canvas

Summary: Shareholders vs. Debtholders

- Having debt holders vs. equity holders in charge could lead to different investment decisions
 - Conflicts of interest lead shareholders to make value destroying investments
 1. Wealth Transfer
 2. Risk Shifting
 3. Debt Overhang
- Conflicts only exist when debt is “risky”. Thus, these are all indirect costs of financial distress.

Bonus: “Other Conflicts”

Management versus Investors

- So far, assumed that managers of the firm will act in the interest of the firm's investors
- But managers may have their own agenda
 - The managers' interests may differ from investors' interests
- This creates conflicts of interest between managers and investors
 - Leads to "Corporate Governance"

Management versus Investors

- Managers may:
 1. Reduce their effort (“shirking”)
 2. Avoid risk (“live the quiet life”)
 3. Divert the firm’s funds (“pet (negative NPV?) projects”)/ outright stealing (less likely)
- Potential Solutions (Corporate Governance):
 - A. Monitoring (board of directors)
 - B. Align incentives using compensation
 - C. Use firm’s capital structure to align incentives

Set A: Monitor Manager's Effort

- Free-rider problem prevents small investors from monitoring
 - Shareholders delegate monitoring to board
- Board of directors supposed to look after shareholders' interests
 - Board hires independent auditors
 - Boards are not just hired to monitor, also provide advice
- Lenders (typically banks) also monitor
 - This helps shareholders too

Set B: Align Incentives

- If we can't monitor managers' effort, how about we reward them for good performance
- Give them incentive-based compensation (pay for performance):
 - Stock, Options, or Bonuses
- Directly aligns managers' incentives with shareholders' incentives
 - Ask management to put in equity
 - Common in small firms; LBOs etc.

Set C: Using Firm's Capital Structure

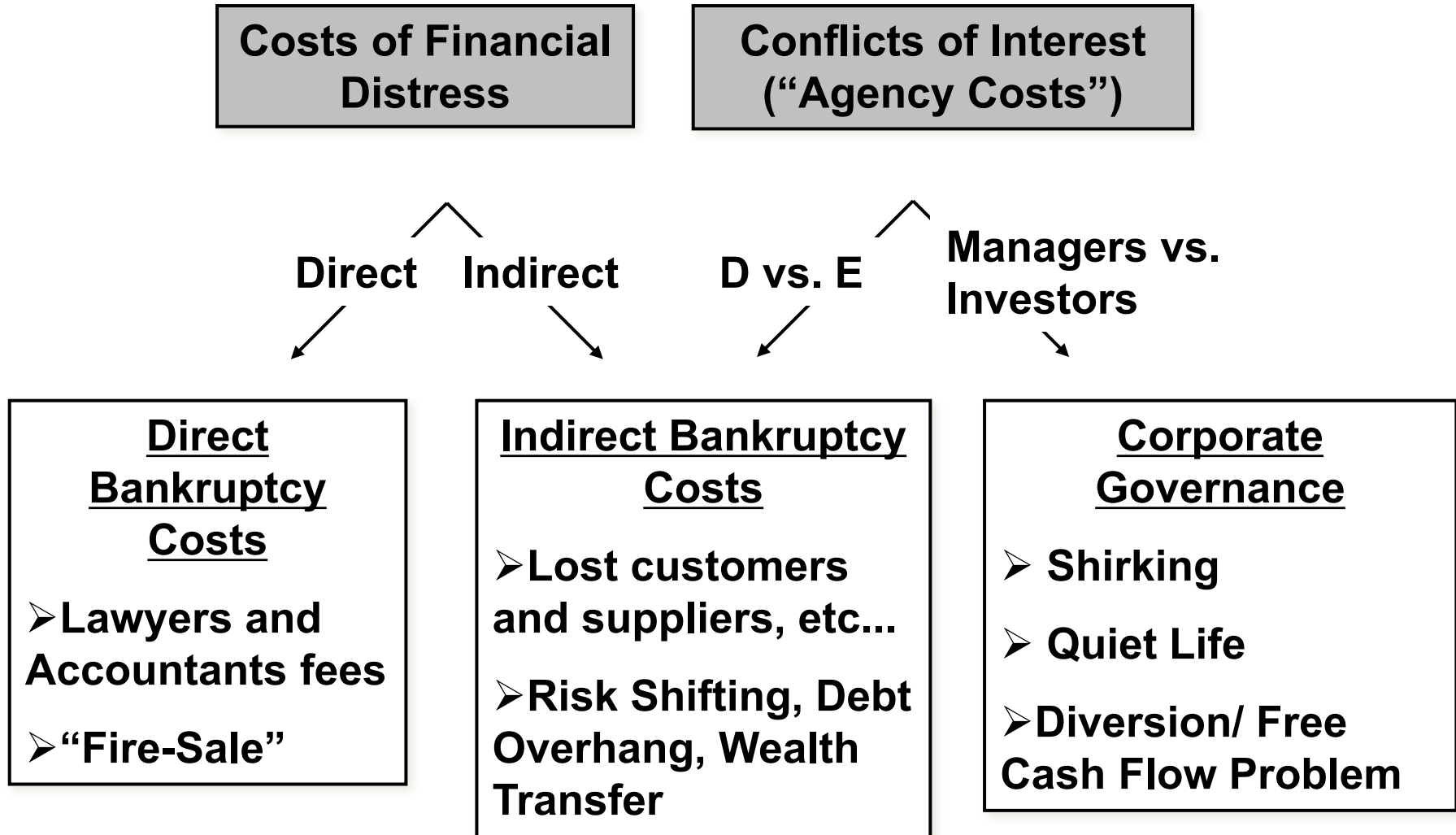
- Monitoring and incentive alignment may be insufficient
 - Difficult to monitor business decisions
- When companies grow beyond what is profitable
 - Reflects the “Free Cash Flow Problem”: Jensen (1986)
- Managers have excess free cash flows but few profitable investment opportunities
 - Managers have inherent need to grow (“empire-building”)
 - Problem more pronounced for mature cash-cow companies
- Debt (or dividends) can help discipline managers
 - Forces them to pay out cash to investors
 - Often implemented through LBO

Wrapping Up

Who Bears The Cost Of Agency Conflicts?

- For DTS and CFD, we saw that the existing shareholders bear the benefit and the cost
- What about agency problems?
 - These arise from shareholders expropriating debtholders (or managers expropriating investors)
- Again, the entire cost is borne by the existing shareholders
 - If debtholders anticipate these problems, they adjust price of debt to reflect the future expropriations

Various Costs in a Snapshot



Takeaways: Agency Conflicts

- Conflicts of interest between debt holders and equity holders
 - Shareholders may make value destroying investments
(1) Wealth transfer (2) Risk-shifting (3) Debt overhang
 - Problems exist when debt is “risky”
- Equity holders bear costs of agency conflicts
 - Assume debt holders anticipate these problems ex-ante and price it in the cost of debt
- Conflicts between managers and investors
 - (1) Shirking (2) Quiet Life (3) Diversion (“empire building”)
 - Corporate governance can help through: (A) Monitoring (B) Incentives (C) Capital structure to reduce “Free Cash Flows”